Lab 3 — Introduction to React.js

Aim

Build a component-based front end using React.

Input / Source Code

App.jsx

```
import { useEffect, useState } from "react";
import TaskList from "./TaskList";
import AddTaskForm from "./AddTaskForm";
function App() {
 const [tasks, setTasks] = useState(() => {
 return JSON.parse(localStorage.getItem("tasks")) || [];
  });
  useEffect(() => {
localStorage.setItem("tasks",JSON.stringify(tasks));
  }, [tasks]);
  const addTask = (task) => {
setTasks([...tasks, task]);
  };
  const deleteTask = (index) => {
setTasks(tasks.filter((_, i) => i !== index));
  };
  return (
    <div>
      <h1>Todo Application</h1>
      <AddTaskForm addTask={addTask} />
      <TaskList tasks={tasks} deleteTask={deleteTask} />
</div>
  );
} export default App;
```

AddTaskForm.jsx

```
import { useState } from "react";
```

```
function AddTaskForm({ addTask }) {
  const [task, setTask] = useState("");
  const [error, setError] = useState("");
  const handleSubmit = () => {
    if (task.trim() === "") {
setError("Please enter a task.");
      return;
    }
    addTask(task);
setTask("");
setError("");
  };
  return (
<div>
 <input
value={task}
       onChange={(e) => setTask(e.target.value)}
placeholder="Enter a task"
     />
     <button onClick={handleSubmit}>Add Task</button>
<p>{error}</p>
    </div>
  );
}
export default AddTaskForm;
```

TaskList.jsx

```
import TaskItem from "./TaskItem";
function TaskList({ tasks, deleteTask }) {
return (
   <ul>
 {tasks.map((task, index) => (
```

```
<TaskItem
  key={index}
 task={task}
   deleteTask={() => deleteTask(index)}
       />
     ))}
   </ul>
  );
}
export default TaskList;
```

TaskItem.jsx

```
function TaskItem({ task, deleteTask }) {
return (
   <li>
     {task}
     <button onClick={deleteTask}>Delete</button>
   </li>
  );
}
export default TaskItem;
```

Output

Initial Screen

Todo Application

[Enter a task] [Add Task]

After Adding Task

Todo Application

[Enter a task] [Add Task]

- Complete React Assignment [Delete]
- Practice React Hooks [Delete]

Validation Output

If the Add Task button is clicked without entering a task:

Please enter a task.

Component Tree

App

├— AddTaskForm

├— TaskList

| ├— TaskItem

| └— TaskItem

└— Local Storage